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Winter, 2026

B. Tech 4th Semester Examination

Hydraulic Engineering- Theory

Course Code: BCE23404T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

1. Choose the correct answer. **(1 x 10 = 10)**

(i) The velocity distribution in laminar flow through a circular pipe follows the
(a) parabolic law
(b) linear law
(c) logarithmic law
(d) none of the above

(ii) For laminar flow through a circular pipe, the maximum velocity is
(a) 1.5 times the average velocity
(b) 2.0 times the average velocity
(c) 0.5 times the average velocity
(d) 2.5 times the average velocity

(iii) Total drag on a body is the sum of
(a) pressure drag and velocity drag
(b) friction drag and velocity drag
(c) pressure drag and friction drag
(d) none of the above

(iv) In pipelines, a surge tank is provided
(a) to relieve the pressure due to water hammer
(b) to provide additional water head
(c) to overflow the pipe line when suddenly closed
(d) to remove the frictional loss in pipe

(v) Which of the following turbine is suitable for specific speed ranging from 300 to 1000 and head below 30 m?

(a) Francis
(b) Kaplan
(c) Propeller
(d) Pelton wheel

(vi) Francis turbine is

(a) an impulse turbine
(b) a reaction turbine
(c) a tangential flow turbine
(d) an axial flow turbine

(vii) For maximum efficiency of a series of curved vanes, the vane speed is

(a) equal to the jet speed
(b) ½ of the jet speed
(c) ¼ of the jet speed
(d) ¾ of the jet speed

(viii) The delivery valve, while starting centrifugal pump, is kept

(a) fully closed
(b) fully open
(c) half open
(d) in any position

(ix) The difference between the theoretical discharge and actual discharge is called

(a) coefficient of discharge
(b) percentage slip
(c) slip
(d) negative slip

(x) The power which appears in the expression for the specific speed is the

(a) shaft power
(b) water power
(c) power into the turbine
(d) none of the above

2. Define laminar flow.A crude oil of viscosity 0.9 Poise and specific gravity 0.8 is flowing through a horizontal circular pipe of diameter 80 mm and of length 15 m. Calculate the difference of pressure at the two ends of the pipe, if 50 kg of the oil is collected in a tank in 15 seconds.

OR(1 + 5 = 6)

Derive the relation for flow of viscous fluid through a circular pipe. Also calculate the ratio of maximum velocity to average velocity.

3. Explain hydrodynamically smooth and rough boundary.

OR(5)

A pipe-line carrying water has average height of irregularities projecting from the surface of the boundary of the pipe as 0.20 mm. what type of the boundary is it? The shear stress development is 7.848 N/m^2. Take value of kinematic viscosity for water as 0.01 stokes.

4. Define Energy Thickness and derive the expression for it.

OR(10)

Find the ratio of (i) displacement thickness to momentum thickness and (ii) momentum thickness to energy thickness for velocity distribution in the laminar boundary layer:

$$\frac{v}{V} = \frac{2y}{\partial} = \frac{y^2}{\partial^2}$$

5. What is equivalent pipe? Derive the Dupit's equation for equivalent pipe.

OR(5)

Find the head loss due to friction in a pipe of diameter 250 mm and length 60 m, through which water is flowing at a velocity of 3 m/s using (i) Darcy's formula and Chezy's formula for which C = 55. Take ϑ for water = 0.01 stoke.

6. A jet of water having a velocity of 30 m/s strikes a curved vane, which is moving with a velocity of 15 m/s. The jet makes an angle of 30° with the direction of motion of vane at inlet and leaves at an angle of 120° to the direction of motion of vane at outlet. Draw the velocity triangles and calculate:

(i) Vane angles, if the water enters and leaves the vane without shock
(ii) Work done per second per unit weight of water striking the vanes per second
(iii) Efficiency

OR(7)

Find the expression for force exerted and work done by a jet of water on a moving curved plate (jet striking at the centre of plate). Also establish the condition for maximum efficiency.

7. Draw the velocity triangles and derive the expression for work done and maximum efficiency by a Pelton Wheel.

OR(7)